



Bicarbonate enhanced removal of triclosan by copper(II) catalyzed Fenton-like reaction in aqueous solution

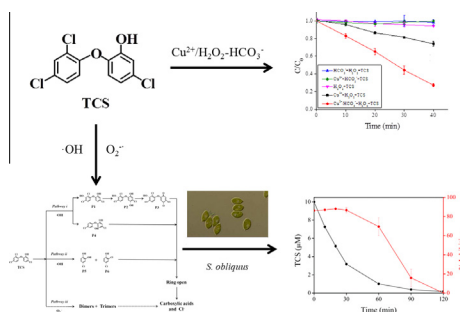
Jianbiao Peng, Huanhuan Shi, Jianhua Li, Lianhong Wang, Zunyao Wang, Shixiang Gao*

State Key Laboratory of Pollution Control and Resource Reuse, School of the Environment, Nanjing University, Nanjing 210023, People's Republic of China

HIGHLIGHTS

- HCO_3^- at low concentration greatly enhanced triclosan removal in $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system.
- $\text{O}_2^{\cdot-}$ and $\cdot\text{OH}$ as the main ROS were responsible for the removal of TCS.
- The degradation pathways for TCS oxidation reaction were proposed.
- Toxicity of TCS solutions was greatly decreased after treatment by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$.

GRAPHICAL ABSTRACT



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ABSTRACT

The antimicrobial and preservative agent triclosan (TCS) is an emerging and persistent pollutant, and is ubiquitous in the environment. Low cost and efficient removing technology for TCS in water has been the target of many researches. In this study, effective removal of TCS from water was achieved by $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system in the presence of environmental relevant concentration of bicarbonate (HCO_3^-). The removal efficiency of TCS increased with the increase of HCO_3^- concentration from 0 to 2.0 mM, while further increase of the HCO_3^- concentration caused the decrease of removal efficiency. TCS (0.01 mM) could be totally removed over 120 min when HCO_3^- was 2.0 mM (pH 8.72) in the presence of 0.01 mM Cu^{2+} ions at 25.0 °C. In order to explore the reaction mechanism, radical scavenging technologies, molecular probe compound *p*-Chlorobenzoic acid (PCBA) and cyclovoltammetric (CV) experiments were employed to determine the reactive oxygen species (ROS), the steady-state concentrations of $\cdot\text{OH}$ and Cu^{2+} -complex, respectively. The results suggested that the Cu^{2+} -complex formed between Cu^{2+} and HCO_3^- could catalyze H_2O_2 to produce $\text{O}_2^{\cdot-}$ and $\cdot\text{OH}$, which were responsible for the removal of TCS. Based on the reaction mechanism and identified intermediates of the oxidation process using TOF–LC–MS techniques, the reaction pathways for the oxidation of TCS were proposed including the cleavage of the ether bond, hydroxylation and polymerization. Finally, evaluation on the toxicity change of TCS reaction solution mediated by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system with the freshwater green alga *Scenedesmus obliquus* showed significant detoxification of TCS.

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1. Introduction

Triclosan (5-chloro-2-[2,4-dichloro-phenoxy]-phenol, TCS) is a synthetic, broad-spectrum antimicrobial agent used in household cleaning products, personal care products, and consuming

* Corresponding author.

E-mail address: ecsxg@nju.edu.cn (S. Gao).

products such as soaps, shampoos, and trash bags [1,2]. Wide-spread use of products containing TCS has resulted in the ubiquitous presence of this compound in the environment. It has been detected at relatively high concentrations in various environmental matrices. For example, TCS concentrations were reported from 0.07 to 14,000 $\mu\text{g/L}$ in wastewaters, from 50 to 2300 ng/L in freshwater streams, from 50 to 150 ng/L in seawater and from 1 to 35 $\mu\text{g/kg}$ in sediments [1,3,4]. Also, TCS was ubiquitous in biotic samples such as fish and human breast milk [5]. More importantly, many researchers have found adverse effects of TCS on aquatic organisms in receiving waters such as acute toxicity, chronic toxicity, and endocrine disrupting effect [6,7].

A number of studies have demonstrated that photolysis, biodegradation, and adsorption processes can effectively remove TCS from water. However, there are several drawbacks regarding these treatment techniques. For instance, TCS can be transformed to form detrimental intermediates such as 2,7/2,8-dichlorodibenzo-*p*-dioxins [8], 2,4-dichlorophenol [9], 2,4,6-trichlorophenol, and chloroform [10] under natural sunlight or UV light in aqueous condition. Low efficiency of biodegradation process towards TCS as well as the formation of more lipophilic and bioaccumulative methyl-triclosan limited the development of the process in TCS removal in waters [2,11]. As for adsorption of TCS on different sorbents such as activated carbon, kaolinite and montmorillonite [12], it is strictly controlled by various kinds of conditions, especially pH. As thus, TCS can easily be desorbed from sorbents once conditions change. In this context, alternative or post-treatment is required after adsorption and biological treatment processes.

Fenton and Fenton-like systems with transition metal ions and hydrogen peroxide are effective and low cost advanced oxidation processes (AOPs) used to remove various kinds of organic contaminants. Of them, $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ systems have been proven effective in removing lignin [13], polycyclic aromatic hydrocarbons [14], as well as synthetic dyes [15–17], and has the advantage of function well at a broad pH range (3–9) [18].

Bicarbonate (HCO_3^-) is one of the most abundant anions at concentrations ranging from 1.0 to 5.0 mM in fresh water [19]. Generally, bicarbonate was thought to have a negative effect on AOPs as it was a radical quencher [20]. However, it was noticed that HCO_3^- also acted as an efficient activator for H_2O_2 to generate reactive oxygen species (ROS) such as peroxyxymonocarbonate (HCO_4^-) and singlet oxygen ($^1\text{O}_2$) [20–23]. Some recent studies suggested that HCO_3^- could accelerate the degradation of pollutants in both homogeneous and heterogeneous AOPs [21]. For example, the $\text{Co}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system and $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system exhibited high catalytic capacity to destroy organic dyes [22,23]. Hence, it is worthy to investigate the role of HCO_3^- in the TCS degradation process catalyzed by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system under mild condition.

In this study, the efficiency of $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system for the degradation of TCS in aqueous solution was evaluated. The influence of Cu^{2+} concentration, reaction temperature, and HCO_3^- concentration on the TCS degradation efficiency were studied. Furthermore, in order to explore the reaction mechanism, radical scavenging technologies, molecular probe compound *p*-Chlorobenzoic acid (PCBA) and cyclic voltammetric (CV) experiments were employed to determine the main ROS, the steady-state concentrations of $\cdot\text{OH}$ and Cu^{2+} -complex, respectively, involved in the oxidation process. Besides, the formed intermediates were identified using TOF-LC-MS. To sum up, the reaction mechanism and the possible degradation pathways of TCS were further proposed according to the experimental results and related literature. Finally, toxicity changes of TCS reaction solutions during oxidation processes were examined using the freshwater unicellular green alga *Scenedesmus obliquus* (*S. obliquus*).

2. Material and methods

2.1. Standards and reagents

HPLC grade TCS (97%) and Catalase (from bovine liver, 1 g) were obtained from Sigma-Aldrich (Shanghai, China). *p*-Chlorobenzoic acid (PCBA, 99%) was purchased from J&K scientific (Shanghai, China). Other chemicals in analytical grade purity were purchased from Nanjing Chemical Reagent Co. (Nanjing, China). Methanol was supplied by Tedia Company (Fairfield, USA). All of the reagents were used without further refinement. All glassware was soaked in diluted HNO_3 solution before use.

All solutions were prepared using $18.25 \text{ M}\Omega\cdot\text{cm}^{-1}$ ultrapure water (UPW). Cu^{2+} stock solution (1 mM) was prepared by dissolving an appropriate amount of $\text{CuCl}_2\cdot 2\text{H}_2\text{O}$ with 10 mM HCl solution. H_2O_2 (100 mM) stock solution was made from a 30% w/w H_2O_2 solution. TCS stock solution (1.0 mM) was prepared in 0.01 mM NaOH solution. Stock solutions were kept in glass bottles covered with foil and refrigerated at 4 °C for storage.

2.2. Experimental procedure

Oxidation reactions were conducted in 15 mL Nessler tubes, which were incubated in a constant temperature rotary incubator shaker at 25.0 ± 0.3 °C, 150 rpm. The reaction was initiated through the addition of H_2O_2 . Reaction solutions (10 mL) were prepared in UPW containing 0.01 mM TCS, 0.01 mM Cu^{2+} , 1.0 mM H_2O_2 and one of five HCO_3^- concentrations ranging from 0 to 10 mM. The solution pH was from 8.63 to 8.97. Identical reactors without Cu^{2+} were used as controls. At prespecified time intervals (0, 10, 20, 30, and 40 min) one set of three replicate tubes were sacrificed and 0.5 mL reaction solution was withdrawn from each tube, which was quenched by 0.5 mL 1.5 mM Na_2SO_3 solutions immediately for the high performance liquid chromatography (HPLC) analysis to measure TCS concentration. In particular, the reactions were quenched by catalase in the product identification and toxicity evaluation.

For the product identification, SPE cartridges packed with LC-C18 (500 mg, 6 mL) from Anpel, Shanghai were used. Samples were performed in triplicate with solutions quenched at different reaction time (0, 10, 20, 30, 60, 90 and 120 min). Cartridges were placed on a vacuum manifold, conditioned with 10 mL methanol and 10 mL UPW, after which 120 mL reaction solution was sampled through. After loading the samples to the cartridges, 10 mL UPW was used to rinse the cartridge. Excess of water remaining on the cartridge was removed by vacuumizing overnight. The compounds retained on the cartridge were desorbed with two portions of 3.0 mL methanol for analysis by liquid chromatography-tandem mass spectrometry (LC-MS).

2.3. Analytical methods

The concentration of TCS in aqueous solution was determined by HPLC equipped with a Synchronis C18 column (250 mm $\times$ 4.6 mm, 5 μm) and an UV detector (220 nm). The mobile phase was a methanol/water mixture (90/10, v/v), the flow rate was set to 0.8 mL/min and the injection volume was 20 μL . The retention time for TCS was 7.6 min. The HPLC analytical method of PCBA is listed in Table S1.

In order to confirm whether Cu(II) -complex exist in $\text{Cu}^{2+}\text{-HCO}_3^-$ aqueous solution, a CHI760E-VA computer, with a Glassy carbon and a Pt plate as working electrode and counter electrode, was used to conduct the cyclic voltammetric (CV) experiments. Prior to the measurements, the working solution was deaerated using high purity N_2 for 15 min. The scan rate was set at 20 mV/s. Potentials

were measured in a 0.1 M NaCl electrolyte solution and were reported vs. a hanging mercury drop electrode.

LC–MS analysis was performed on a TOF 5600 (AB Sciex, Foster City, CA), equipped with an Agilent (Palo Alto, CA) 1200 series HPLC system. A Zorbax Eclipse Plus C18 column (150 × 2.1 mm, 5 μm particle-size) from Agilent was used for the separation of the analytes. The mobile phase was run with an isocratic program of 90% methanol and 10% water (with 5% methanol, v/v). The elution flow rate was 0.2 mL/min and the injection volume was 5 μL. A TOF MS scan type in negative electrospray ionization (ESI[−]) mode over the range m/z = 50–1200 was used to analyze TCS and its oxidation products under optimized conditions. The optimized mass parameters were: ion source temperature 550 °C, ion spray voltage −4500 V, nebulizer gas 55 psi, auxiliary gas 55 psi, curtain gas 35 psi.

Details of ion chromatograph (IC) and total organic carbon (TOC) analysis could be found in analytical methods in [Supplementary Material](#).

2.4. Toxicity evaluation

A freshwater unicellular green alga *S. obliquus*, which was found to be very sensitive to TCS and easily cultured [24–26], was used to test toxicity changes during the oxidation of TCS by Cu^{2+} - HCO_3^- system in aqueous solution with H_2O_2 . *S. obliquus* was purchased from the Institute of Hydrobiology, Chinese Academy of Sciences (Wuhan, China) and subsequently cultured using SE medium (OECD Guideline 201 for the Testing of Chemicals, Alga, Growth Inhibition Test, 1984). The concentration of *S. obliquus* was determined in two ways: direct counting under the microscope using the haemocytometer and measurement of the optical density by using a spectrophotometer at 650 nm (OD_{650}).

The test organism was in the logarithmic phase when introduced to the reaction solution and the standard algal culture medium. Tests were performed in triplicate with solutions sampled at different reaction time (0, 10, 20, 30, 60, 90 and 120 min). The identical solutions without TCS were used as controls. Test vessels were maintained at 25.0 ± 0.5 °C, with a 12:12 h light/dark cycle. The initial OD_{650} of *S. obliquus* was adjusted to 0.04 or so after inoculation. The *S. obliquus* OD_{650} were measured at the beginning of the test and every 24 h until 96 h except for the first 12 h. Toxicities of the reaction solution was evaluated by the growth inhibition of *S. obliquus*.

3. Results and discussion

3.1. Catalytic activity of $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- system towards TCS degradation

The removal of TCS under different reaction conditions was displayed in Fig. 1. No significant decrease of TCS concentration was observed in several control experiments including H_2O_2 , H_2O_2 - HCO_3^- and Cu^{2+} - HCO_3^- systems, while apparent removal of TCS was found in both systems with $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ and $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- , and the removal efficiency of $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- system was higher than that of $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system. Furthermore, instead of HCO_3^- , common anions NO_3^- , CO_3^{2-} and OH^- did not give rise to the observed degradation of TCS (Fig. S1). It was not surprising that the degradation rate of TCS decreased with the decrease of Cu^{2+} ion concentrations (Fig. S2A). Nevertheless, it was more intriguing that nearly complete degradation of TCS was observed after 120 min (as shown later in Fig. 7) even at a very low concentration of 0.01 mM Cu^{2+} ions, which is within the threshold of the national emission standard for copper in wastewater (lower than 1 mg/L) in China (GB 25467-2010). The concentration was also lower than

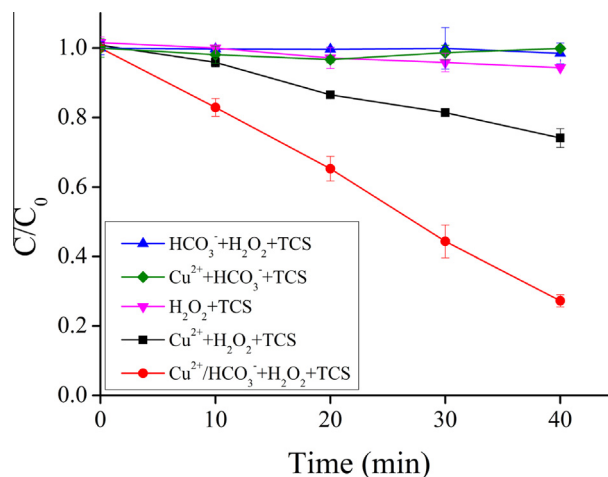


Fig. 1. Time evolution of TCS concentration under different conditions. Conditions: Cu^{2+} 0.01 mM, HCO_3^- 2 mM, H_2O_2 1 mM, TCS 0.01 mM, 25 °C. Error bars represent standard deviations ($n = 3$).

other copper-based Fenton-like systems in the previous literatures. For instance, 85–95% decolorization of Remazol Brilliant Blue R (RBBR), Poly B-411, and Chicago Sky Blue required 10 mM Cu^{2+} ions [27]. The 0.1 mM $\text{Cu(II)}/\text{H}_2\text{O}_2$ system exhibited high removal for Methanol, Reactive Black 5 and Arsenite under neutral or alkaline pH conditions [28].

The TCS degradation efficiency was also significantly affected by the reaction temperature. As shown in Fig. S2B, the higher the reaction temperature, the faster the TCS degradation. The observed degradation rate constants (k_{obs}) were 1.27×10^{-2} , 3.23×10^{-2} , 4.51×10^{-2} , and $8.41 \times 10^{-2} \text{ min}^{-1}$ at 20, 25, 30, and 40 °C, respectively. Moreover, the apparent activation energy (E_a) of $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- system for the catalyzed TCS oxidative degradation was estimated to be 68 kJ mol^{-1} , which was calculated using the Arrhenius equation with the experimental data. Thus, it can be concluded that higher reaction temperature was conducive to the removal efficiency.

3.2. Reaction mechanism

Two major alternative mechanisms, i.e. the “classical free radical” mechanism and the “complex” mechanism, accounted for the reaction between copper and H_2O_2 . Specially, $\cdot\text{OH}$, $^1\text{O}_2$ and O_2^- were usually identified as the ROS during the reaction in the “classical free radical” mechanism [29,30]. The “complex” mechanism involved the complexes of copper formed with peroxide or some other ligands [15,31]. Recent studies, however, have indicated that the two pathways might be compatible [32].

To determine the ROS involved in TCS reaction in $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- system, 2-propanol, sodium azide (NaN_3), and nitroblue tetrazolium (NBT) were used as $\cdot\text{OH}$ scavenger, $^1\text{O}_2$ quencher, and O_2^- indicator, respectively. As reported in previous work [33–35], the concentrations of the scavengers employed here were sufficient to inhibit $\cdot\text{OH}$, $^1\text{O}_2$ and O_2^- . Oxidative degradation kinetics of TCS in the presence of various scavengers was shown in Fig. 2.

As can be seen in Fig. 2A, about 70.8% of inhibition efficiency on the TCS degradation occurred with the addition of 2-propanol (65.7 mM), which suggested that $\cdot\text{OH}$ was participated in the reaction of TCS under the experimental conditions. Similarly, the degradation of TCS was inhibited by 92.4% when NBT (0.30 mM) were added in the reaction system (Fig. 2B), which confirmed the significant role of O_2^- in the TCS oxidative process. In addition, no obvious inhibitory effect was observed with the addition of NaN_3 (0.30 or 0.03 mM) into the reaction mixture (Fig. 2C), implying that

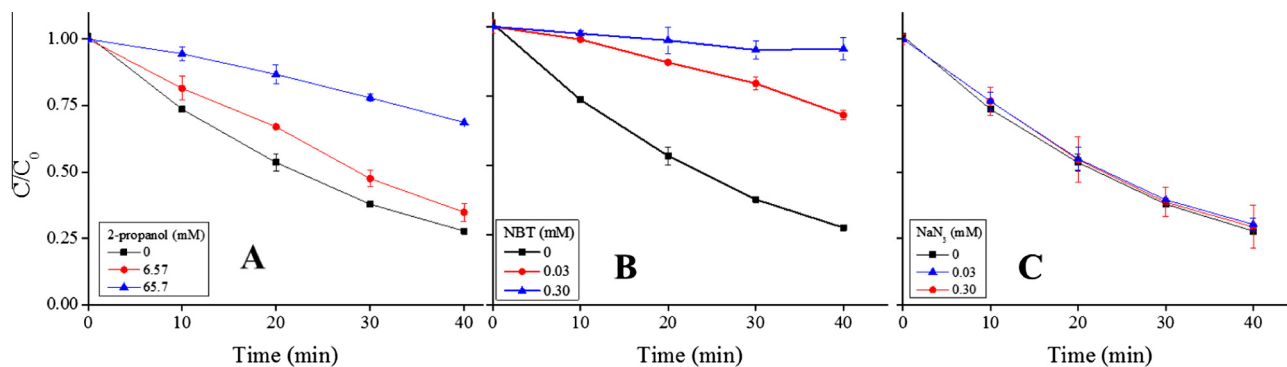


Fig. 2. Effect of (A) 2-propanol, (B) NBT and (C) NaN_3 on TCS degradation by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system. Conditions: Cu^{2+} 0.01 mM, HCO_3^- 2 mM, H_2O_2 1 mM, TCS 0.01 mM, 25 °C, 2-propanol 6.57 and 65.7 mM, NBT 0.03 and 0.30 mM, NaN_3 0.03 and 0.30 mM. Error bars represent standard deviations ($n = 3$).

the contribution of $^1\text{O}_2$ to the degradation of TCS in the $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system was negligible.

On the basis of the conclusion that HCO_3^- enhanced removal of TCS in the $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system as mentioned in Section 3.1, the effect of HCO_3^- concentrations (0, 1.0, 2.0, 5.0 and 10 mM) on TCS degradation was further investigated (Fig. 3). It was important to note that HCO_3^- was the main speciation accounting for 96.4% of $\text{H}_2\text{CO}_3\text{-HCO}_3^-\text{-CO}_3^{2-}$ system in this study, which was calculated with the Visual MINTEQ computer program based on the alkalinity and pH conditions (Fig. S3). As shown in Fig. 3, the removal efficiency of TCS increased with the increase of HCO_3^- concentration from 0 to 2.0 mM. When HCO_3^- was 2.0 mM, the removal efficiency of TCS reached to 73.0% over 40 min. However, further increase of the HCO_3^- concentration caused the decrease of TCS removal efficiency. It has been reported that HCO_3^- was a $\cdot\text{OH}$ radical scavenger [28]. The decrease of removal efficiency of TCS at high concentrations of HCO_3^- maybe resulted from the scavenging effect of excess HCO_3^- on $\cdot\text{OH}$ in the $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system. Hence, the concentration of HCO_3^- is critical to the copper-based Fenton-like system regarding TCS removal.

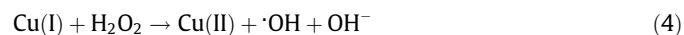
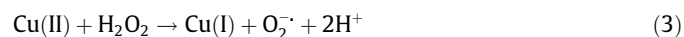
Molecular probe technology has been proved to be reliable and useful in environmental photochemistry studies [36]. In this study, we employed PCBA as a probe, and determined the steady-state concentration of $\cdot\text{OH}$ ($[\cdot\text{OH}]_{ss}$) in $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ and $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system. Fig. S4 showed the loss of PCBA (20 μM) oxidized by $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ with and without HCO_3^- . The pseudo-first-order rate constants $k_{obs,PCBA}$ such obtained were $1.40 \times 10^{-2} \text{ min}^{-1}$ and

$0.86 \times 10^{-3} \text{ min}^{-1}$ for $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ and $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system, respectively. In light of PCBA scavenged $\cdot\text{OH}$ with a second-order rate constant of $k_{OH,PCBA} = 5.20 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$ [37], $[\cdot\text{OH}]_{ss}$ in $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ and $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system were calculated as $4.49 \times 10^{-14} \text{ M}$ and $2.76 \times 10^{-15} \text{ M}$, respectively, according to the following equations (Reactions (1) and (2)). $[\cdot\text{OH}]_{ss}$ in $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system were an order of magnitude higher than that of $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ system, which could account for the enhanced removal of TCS by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system



$$-d[\text{PCBA}]/dt = k_{obs,PCBA}[\text{PCBA}] = k_{OH,PCBA}[\text{PCBA}][\cdot\text{OH}]_{ss} \quad (2)$$

Generally, it has been proposed that the decomposition of H_2O_2 , catalyzed by Cu(II)/Cu(I) , produces ROS such as O_2^- and $\cdot\text{OH}$ (Reactions (3) and (4)) [38,39]:



Therefore, the valence change of Cu(II) during TCS oxidation by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system was studied. As stated above, the collected UV-vis spectra of reaction solutions in the system were also used to examine apparent absorbance at 520 nm where Cu(I) has maximum absorption [23,40]. However, no significant absorbance at 520 nm was observed with the increase of Cu^{2+} concentration, even when the Cu^{2+} concentration elevated to 0.5 and 1.0 mM (Fig. S5). Thus, the valence of Cu(II) change to Cu(I) did not take place in $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system, which excluded the production of O_2^- and $\cdot\text{OH}$ from the Reactions (3) and (4).

In the literature [41], Luo and co-workers reported that the Co^{2+} -complex formed between Co^{2+} and HCO_3^- could catalyze H_2O_2 to produce $\cdot\text{OH}$ and the presence of Co^{2+} -complex was confirmed by the absorbance change at 260 nm, where Co^{2+} -complex has maximum absorption. Hence, in order to have a better understanding of the reaction mechanism, the contribution of Cu^{2+} -complex to the TCS oxidation was determined. As can be seen in Fig. 4A, a significant increase of absorbance at 270 nm was observed after the addition of HCO_3^- into the Cu^{2+} or $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ solution, which implied that Cu^{2+} -complex was formed in the $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system. This statement was further confirmed by CV measurements at different HCO_3^- concentrations (0, 1.0, 2.0, 5.0 and 10 mM), and the voltage-current curves were shown in Fig. 4B. Because HCO_3^- was a negatively charged ligand, the formed Cu^{2+} -complex in $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system was expected to have a more negative reduction potential similar to the $\text{Co}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system [22]. As expected, the peak potential in the $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ solutions shifted to more negative potentials

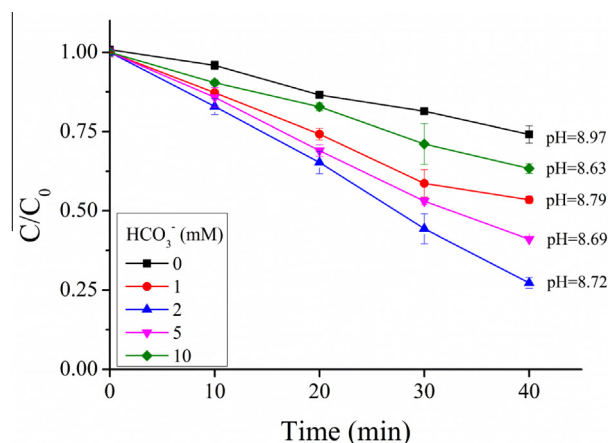


Fig. 3. Effect of HCO_3^- concentration on TCS removal by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system. Conditions: Cu^{2+} 0.01 mM, H_2O_2 1 mM, TCS 0.01 mM, 25 °C. Error bars represent standard deviations ($n = 3$).

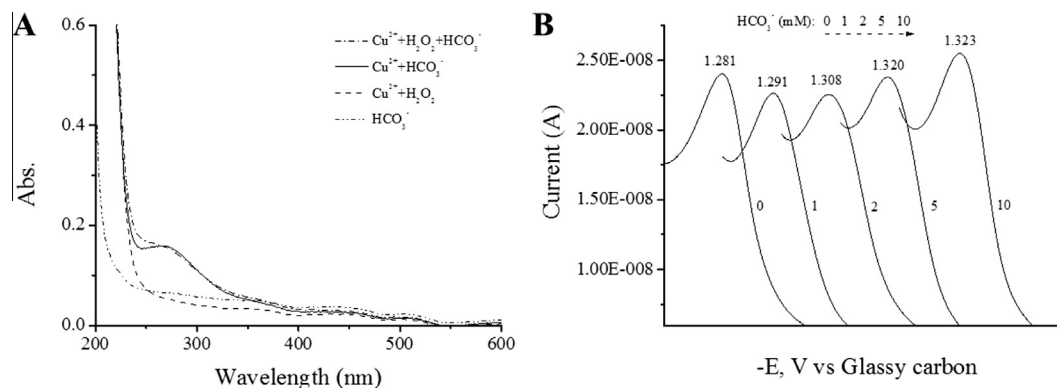


Fig. 4. (A) UV-vis spectra of varying reaction solutions under different conditions: Cu^{2+} 0.01 mM, HCO_3^- 2 mM, H_2O_2 1 mM, 25 °C; (B) Voltage-current curves of Cu^{2+} reduction in aqueous solution of 0.1 NaCl at different concentrations of HCO_3^- : Cu^{2+} 0.01 mM, 25 °C; the curves shifted on the x axis for clarity and the peak potentials were indicated at the top of each curve.

was found, which reinforced the conclusion about the formation of Cu^{2+} -complex between Cu^{2+} and HCO_3^- . As for the number of HCO_3^- coordinated to Cu^{2+} , the various slopes of the peak potential as a logarithmic function of HCO_3^- concentration were plotted according to the literature (Fig. S6A) [22]. A slope of 0.0317 V/log[HCO_3^-] was found in solution at 1.0–10 mM of HCO_3^- concentration which was similar to the slope of 0.0332 V/log[HCO_3^-] in $\text{Co}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- system at HCO_3^- concentrations (5.0–10 mM), indicating that $[\text{Cu}(\text{HCO}_3)]^+$ was the dominant species [22]. In addition, from the result of CV experiment in the presence of 2.0 mM HCO_3^- (Fig. S6B), we could conclude that the reaction is irreversible because of the asymmetric curve up and down.

The role of bicarbonate was further verified by the decomposition experiments of H_2O_2 . The decomposition curves of H_2O_2 by Cu^{2+} alone, $\text{Cu}^{2+}/\text{HCO}_3^-$ and $\text{Cu}^{2+}/\text{HCO}_3^-/\text{TCS}$ systems were displayed in Fig. S7. For Cu^{2+} alone system, approximately no H_2O_2 was decomposed after 40 min reaction, indicating that the free Cu^{2+} is relatively unreactive with H_2O_2 . For $\text{Cu}^{2+}/\text{HCO}_3^-/\text{TCS}$ and $\text{Cu}^{2+}/\text{HCO}_3^-$ systems, approximately 18.1% and 78.3% of the initial H_2O_2 was decomposed after 40 min. The possible reason is the formation of $[\text{Cu}(\text{HCO}_3)(\text{TCS})]^+$ between TCS and $[\text{Cu}(\text{HCO}_3)]^+$, which lead to the decrease in number of coordinated H_2O_2 and its disproportionation rate [23].

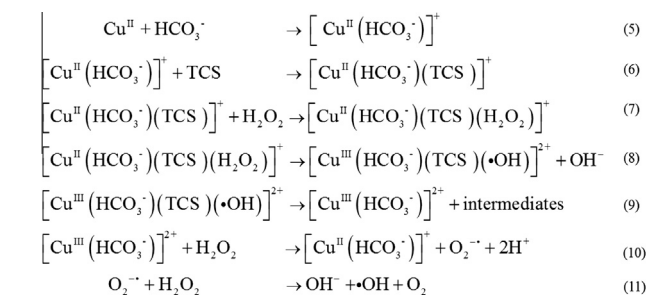
In conclusion, according to the aforementioned results and the related literature [22], a catalysis mechanism for TCS degradation by $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- system was proposed in Scheme 1. In the first step, $[\text{Cu}(\text{HCO}_3)]^+$ was formed between Cu^{2+} and HCO_3^- . Next, $[\text{Cu}(\text{HCO}_3)(\text{TCS})]^+$ was produced due to the coordination of TCS to $[\text{Cu}(\text{HCO}_3)]^+$. When H_2O_2 was present in the solution, the $[\text{Cu}(\text{HCO}_3)(\text{TCS})]^+$ may react directly with it and served as the catalyst to activate H_2O_2 to $\cdot\text{OH}$ and $\text{O}_2^{\cdot-}$ radicals. Hence, we concluded that the “classical free radical” mechanism and the “complex” mechanism

were compatible in this study. The conclusion was different from a previous literature, in which Cheng and co-workers have provided evidence supporting that CuCO_3 was more active than other copper species by the formation of Cu(III) as the main reactive species which were responsible for the dye decolorization [23]. The discrepancy can be resulted from the concentration of HCO_3^- , which was reported both as a $\cdot\text{OH}$ radical scavenger at high concentration and as an efficient activator for H_2O_2 to generate ROS at low concentration. Thus, different concentration of HCO_3^- used lead to the difference of reaction mechanism between aforementioned study (10 mM) and our study (2.0 mM).

3.3. Oxidation products of TCS

A total of eight compounds including six major intermediate products were detected using TOF-LC-MS after SPE procedure. These products were identified based on several important parameters including retention time, mass accuracy, isotope pattern matching, MS/MS spectra and authentic sample as for 2,4-dichlorophenol. Six major intermediates (P1–P6) formed during the oxidative degradation of TCS were listed in Table 1 with their supposed structures, while the ESI[−] mass spectra of them were depicted in Fig. S8. Of them, three products P1, P2 and P3 with isotopic patterns of three chlorine atoms and $[\text{M}^-]$ at m/z 302.9381, 318.9326 and 316.9183, respectively, were detected. These products were designated as hydroxylated products and this hydroxylation mechanism could be found in previous studies [42–46]. P4 was identified as a dechlorinated and $-\text{OH}$ addition derivative of TCS with $[\text{M}^-]$ at m/z 268.9755 presenting an isotopic pattern of two chlorine atoms. Two other intermediates P5 and P6 with a single aromatic ring were detected $[\text{M}^-]$ at m/z 142.9910 and 160.9552, respectively. P5 and P6 were proposed to be generated after the cleavage of the ether bond. Furthermore, the molecular weights of other products at m/z 574.8955, corresponding to the coupling of two TCS molecules after losing two hydrogen atoms, matched the dimers of TCS (Fig. S9). In addition to dimers of TCS, trace amount of trimers of TCS with $[\text{M}^-]$ at m/z 864.8619 was also detected. The formation of dimers and trimers of TCS may be attributed to the preferable attack of $\text{O}_2^{\cdot-}$ on the aromatic rings [47]. Notably, the formation of toxic dioxin derivatives was not detected under the applied experimental conditions.

As can be seen in Fig. 5, the areas of P1–P6 first increased to reach their individual peak value and then gradually decreased. Meanwhile, compared with P1 and P2, P3 reached a maximum concentration after 30 min, presumably P3 maybe the further hydroxylated and oxidized product of P1 and P2 (Fig. 5A). Time evolution profiles of P4, P5 and P6 were shown in Fig. 5B.

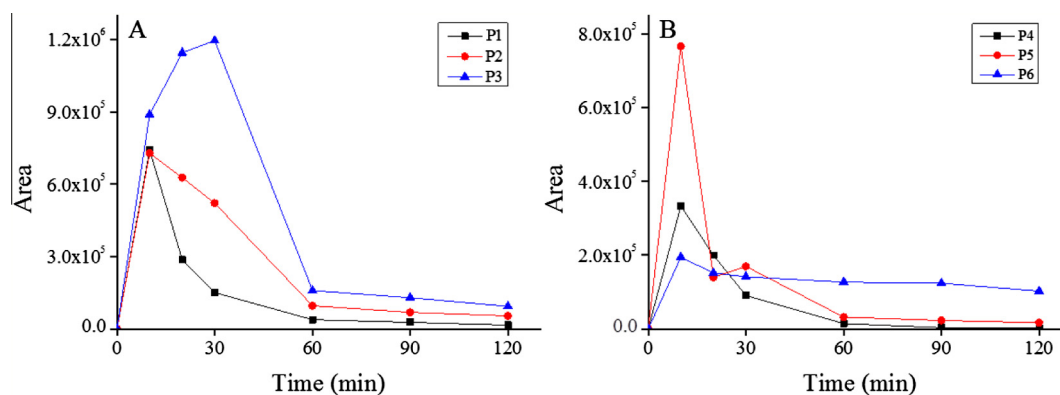


Scheme 1. Proposed reaction mechanism for TCS degradation catalyzed by $\text{Cu}^{2+}/\text{H}_2\text{O}_2$ - HCO_3^- system.

Table 1

Tentative identification of intermediate products (P1–P6) formed during the oxidative degradation of TCS using TOF–LC–MS.

Compound	Suggested structure	Retention time (min)	Formula	Calculated mass	Observed mass	Mass diff. (ppm)
TCS		3.592	C ₁₂ H ₇ O ₂ Cl ₃	286.9439	286.9441	0.70
P1		2.943	C ₁₂ H ₇ O ₃ Cl ₃	302.9388	302.9381	2.30
P2		2.003	C ₁₂ H ₇ O ₄ Cl ₃	318.9337	318.9326	3.40
P3		2.029	C ₁₂ H ₅ O ₄ Cl ₃	316.9181	316.9183	0.60
P4		2.705	C ₁₂ H ₈ O ₃ Cl ₂	268.9777	268.9755	-8.20
P5		2.194	C ₆ H ₅ O ₂ Cl	142.9905	142.9910	3.50
P6		2.583	C ₆ H ₄ OCl ₂	160.9566	160.9552	-8.70

**Fig. 5.** Time evolution profiles of major transformed products (A) P1, P2, P3 and (B) P4, P5, P6 during the oxidative treatment of TCS.

Previous literatures have reported that further oxidation of intermediates usually lead to ring opening, whereby generating a series of carboxylic acids with low molecular weight before complete mineralization, such as glyoxylic, maleic, oxalic, formic, and acetic acid [48]. Therefore, we also investigated whether such carboxylic acids produced or not, formic and acetic acids were detected as shown in Fig. S10A. Finally, the variation of TOC content was measured during the oxidative treatment, and the TOC removal after 120 min was 34.5% which was displayed in Fig. S10B.

Based on scavenging studies and intermediates identification, the possible degradation pathways for the oxidation of TCS by Cu²⁺/H₂O₂-HCO₃⁻ system were proposed in Fig. 6, which mainly involved (i) cleavage of the ether bond through ·OH radical attack (ii) single or multiple hydroxylation of TCS via ·OH radical attack and (iii) dimerization or polymerization of TCS likely via O₂⁻ attack.

3.4. Toxicity evaluation of the TCS reaction solution treated by Cu²⁺/H₂O₂-HCO₃⁻ system

Since it is an essential goal of the treatment process to convert contaminants into innocuous products, experiments were conducted to evaluate the toxicity changes of the reaction mixture via growth toxicity assay of the algae *S. obliquus*. Acute toxicity (expressed as percentage inhibition of the 96 h growth) of different reaction solution samples was depicted in Fig. 7. Notably, the Cu²⁺/H₂O₂-HCO₃⁻ concentration used in this study did not appear to inhibit *S. obliquus* growth (Fig. S11A). The TCS reaction solution treated over the first 30 min treatment showed an 87.0–89.0% inhibition rate on *S. obliquus* growth, and then inhibition percentage sharply decreased. Complete detoxification of the reaction solution was achieved after 120 min treatment. Comparing the algal growth inhibition curve with the residual concentration of

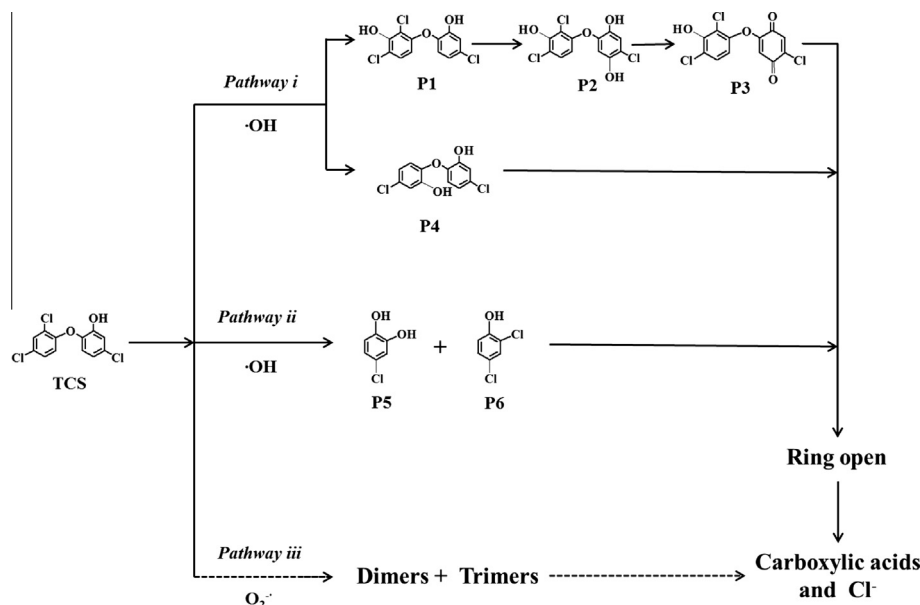


Fig. 6. The proposed pathways for TCS oxidative degradation by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system.

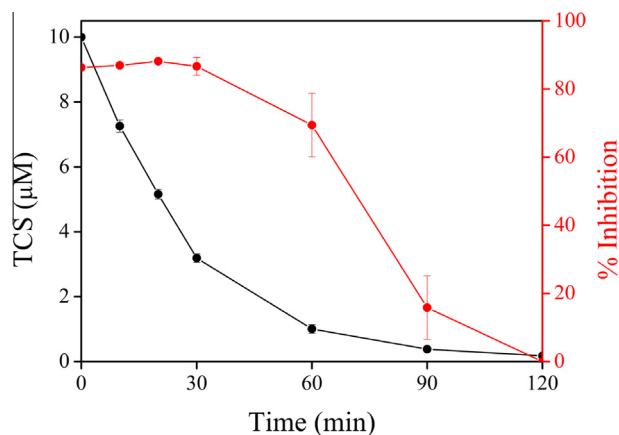


Fig. 7. Inhibition (%) of the 96 h growth of *Scenedesmus obliquus* as a function of treatment time. Error bars represent standard deviations ($n = 3$).

TCS in the solution as reaction programmed (Fig. S11B), the residual TCS in the solution was the major factor determining the toxicity of the treated samples, while the contribution of reaction products to the toxicity was negligible.

4. Conclusion

The present work demonstrated that $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ was an efficient system for the removal of TCS in aqueous solution. Nearly complete degradation of TCS was obtained after 120 min in 2.0 mM HCO_3^- (pH 8.72) solution in the presence of 0.01 mM Cu^{2+} ions at 25.0 °C. According to the results of radical scavenging assays, PCBA and CV experiments, $\cdot\text{OH}$ and $\text{O}_2^{\cdot-}$ produced from the decomposition of H_2O_2 catalyzed by the $[\text{Cu}(\text{HCO}_3)(\text{TCS})]^+$ were responsible for the TCS degradation. Based on the intermediates identified by TOF-LC-MS, the reaction pathways for the degradation of TCS by $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ system involved the cleavage of the ether bond, hydroxylation and polymerization. More importantly, the degradation process resulted in a significant decrease in algal toxicity of TCS according to the growth inhibition tests with *S. obliquus*. In conclusion, $\text{Cu}^{2+}/\text{H}_2\text{O}_2\text{-HCO}_3^-$ is an effective, low cost, and environ-

mentally friendly system, which can be a promising method to supplement the existing treatment techniques of TCS in wastewater treatment plants.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.cej.2016.07.088>.

References

- [1] H. Singer, S. Muller, C. Tixier, L. Pillonel, Triclosan: occurrence and fate of a widely used biocide in the aquatic environment: field measurements in wastewater treatment plants, surface waters, and lake sediments, *Environ. Sci. Technol.* 36 (2002) 4998–5004.
- [2] G.G. Ying, X.Y. Yu, R.S. Kookana, Biological degradation of tricloran and triclosan in a soil under aerobic and anaerobic conditions and comparison with environmental fate modelling, *Environ. Pollut.* 150 (2007) 300–305.
- [3] R.U. Halden, D.H. Paull, Co-occurrence of tricloran and triclosan in U.S. water resources, *Environ. Sci. Technol.* 39 (2005) 1420–1426.
- [4] M.G. Pintado-Herrera, E. Gonzalez-Mazo, P.A. Lara-Martin, Determining the distribution of triclosan and methyl triclosan in estuarine settings, *Chemosphere* 95 (2014) 478–485.
- [5] M. Adolfsson-Erici, M. Pettersson, J. Parkkonen, J. Sturve, Triclosan, a commonly used bactericide found in human milk and in the aquatic environment in Sweden, *Chemosphere* 46 (2002) 1485–1489.
- [6] R. Oliveira, I. Domingues, C. Koppe Grisolia, A.M. Soares, Effects of triclosan on zebrafish early-life stages and adults, *Environ. Sci. Pollut. Res.* 16 (2009) 679–688.
- [7] N. Veldhoen, R.C. Skirrow, H. Osachoff, H. Wigmore, D.J. Clapson, M.P. Gunderson, G. Van Aggelen, C.C. Helbing, The bactericidal agent triclosan modulates thyroid hormone-associated gene expression and disrupts postembryonic anuran development, *Aquat. Toxicol.* 80 (2006) 217–227.
- [8] K. Aranami, J.W. Readman, Photolytic degradation of triclosan in freshwater and seawater, *Chemosphere* 66 (2007) 1052–1056.
- [9] D.E. Latch, J.L. Packer, B.L. Stender, J. VanOverbeke, W.A. Arnold, K. McNeill, Aqueous photochemistry of triclosan: formation of 2,4-dichlorophenol, 2,8-dichlorodibenzo-p-dioxin, and oligomerization products, *Environ. Toxicol. Chem.* 24 (2005) 517–525.

- [10] E.M. Fiss, K.L. Rule, P.J. Vikesland, Formation of chloroform and other chlorinated byproducts by chlorination of triclosan-containing antibacterial products, *Environ. Sci. Technol.* 41 (2007) 2387–2394.
- [11] A. Lindstrom, I.J. Buerge, T. Poiger, P.A. Bergqvist, M.D. Muller, H.R. Buser, Occurrence and environmental behavior of the bactericide triclosan and its methyl derivative in surface waters and in wastewater, *Environ. Sci. Technol.* 36 (2002) 2322–2329.
- [12] S.K. Behera, S.Y. Oh, H.S. Park, Sorption of triclosan onto activated carbon, kaolinite and montmorillonite: effects of pH, ionic strength, and humic acid, *J. Hazard. Mater.* 179 (2010) 684–691.
- [13] T. Watanabe, K.K. Fau, K. Messner, Copper-dependent depolymerization of lignin in the presence of fungal metabolite, pyridine, *J. Biotechnol.* 62 (1998) 221–230.
- [14] J. Gabriel, V. Shah, K. Nesměrák, P. Baldrian, F. Nerud, Degradation of polycyclic aromatic hydrocarbons by the copper(II)-hydrogen peroxide system, *Folia Microbiol.* 45 (2000) 573–575.
- [15] M.H. Robbins, R.S. Drago, Activation of hydrogen peroxide for oxidation by copper(II) complexes, *J. Catal.* 170 (1997) 295–303.
- [16] I.A. Salem, Kinetics of the oxidative color removal and degradation of bromophenol blue with hydrogen peroxide catalyzed by copper(II)-supported alumina and zirconia, *Appl. Catal. B* 28 (2000) 153–162.
- [17] P. Verma, J. Gabriel, T. Trnka, F. Nerud, Copper-ligand complex for the decolorization of synthetic dyes, *Chemosphere* 57 (2004) 1207–1211.
- [18] F. Nerud, P. Baldrian, J. Gabriel, D. Ogbeifun, Decolorization of synthetic dyes by the Fenton reagent and the Cu/pyridine/H₂O₂ system, *Chemosphere* 44 (2001) 957–961.
- [19] M.J. Stiff, Copper/bicarbonate equilibria in solutions of bicarbonate ion at concentrations similar to those found in natural water, *Water Res.* 5 (1971) 171–176.
- [20] L. Zhou, W. Song, Z. Chen, G. Yin, Degradation of organic pollutants in wastewater by bicarbonate-activated hydrogen peroxide with a supported cobalt catalyst, *Environ. Sci. Technol.* 47 (2013) 3833–3839.
- [21] C.S. Liu, K. Shih, C.X. Sun, F. Wang, Oxidative degradation of propachlor by ferrous and copper ion activated persulfate, *Sci. Total Environ.* 416 (2012) 507–512.
- [22] A. Xu, X. Li, S. Ye, G. Yin, Q. Zeng, Catalyzed oxidative degradation of methylene blue by in situ generated cobalt(II)-bicarbonate complexes with hydrogen peroxide, *Appl. Catal. B* 102 (2011) 37–43.
- [23] L. Cheng, M. Wei, L. Huang, F. Pan, D. Xia, X. Li, A. Xu, Efficient H₂O₂ oxidation of organic dyes catalyzed by simple copper(II) ions in bicarbonate aqueous solution, *Ind. Eng. Chem. Res.* 53 (2014) 3478–3485.
- [24] D.R. Orvos, D.J. Versteeg, J. Inauen, M. Capdevielle, A. Rothenstein, V. Cunningham, Aquatic toxicity of triclosan, *Environ. Toxicol. Chem.* 21 (2002) 1338–1349.
- [25] S. Franz, R. Altenburger, H. Heilmeyer, M. Schmitt-Jansen, What contributes to the sensitivity of microalgae to triclosan?, *Aquat. Toxicol.* 90 (2008) 102–108.
- [26] L.H. Yang, G.G. Ying, H.C. Su, J.L. Stauber, M.S. Adams, M.T. Binet, Growth inhibiting effects of 12 antibacterial agents and their mixtures on the freshwater microalga *Pseudokirchneriella subcapitata*, *Environ. Toxicol. Chem.* 27 (2008) 1201–1208.
- [27] M.V. Kirillova, Y.N. Kozlov, L.S. Shul'pina, O.Y. Lyakin, A.M. Kirillov, E.P. Talsi, A. J.L. Pombeiro, G.B. Shul'pin, Decolorization of dyes with copper(II)/organic acid/hydrogen peroxide systems, *Appl. Catal. B* 268 (2003) 26–38.
- [28] H. Lee, H.J. Lee, D.L. Sedlak, C. Lee, pH-dependent reactivity of oxidants formed by iron and copper-catalyzed decomposition of hydrogen peroxide, *Chemosphere* 92 (2013) 652–658.
- [29] J.W. Moffett, R.G. Zika, Reaction kinetics of hydrogen peroxide with copper and iron in seawater, *Environ. Sci. Technol.* 21 (1987) 804–810.
- [30] F.J. Millero, V.K. Sharma, B. Karn, The rate of reduction of copper(II) with hydrogen peroxide in seawater, *Mar. Chem.* 36 (1991) 71–83.
- [31] H. Sigel, C. Flierl, R. Griesser, Metal ions and hydrogen peroxide. XX. On the kinetics and mechanism of the decomposition of hydrogen peroxide, catalyzed by the Cu²⁺-2,2'-bipyridyl complex, *J. Am. Chem. Soc.* 91 (1969) 1061–1064.
- [32] T. Lin, C. Wu, Activation of hydrogen peroxide in copper(II)/amino acid/H₂O₂ systems: effects of pH and copper speciation, *J. Catal.* 232 (2005) 117–126.
- [33] S. Kim, W. Choi, Kinetics and mechanisms of photocatalytic degradation of (CH₃)_nNH_{4-n}⁺ (0 ≤ n ≤ 4) in TiO₂ suspension: the role of OH radicals, *Environ. Sci. Technol.* 36 (2002) 2019–2025.
- [34] S.X. Liang, L.X. Zhao, B.T. Zhang, J.M. Lin, Experimental studies on the chemiluminescence reaction mechanism of carbonate/bicarbonate and hydrogen peroxide in the presence of cobalt(II), *J. Phys. Chem. A* 112 (2008) 618–623.
- [35] R.h. Liu, S.y. Fu, H.y. Zhan, L.A. Lucia, General spectroscopic protocol to obtain the concentration of the superoxide anion radical, *Ind. Eng. Chem. Res.* 48 (2009) 9331–9334.
- [36] W.R. Haag, J. Hoigne, Singlet oxygen in surface waters. 3. Photochemical formation and steady-state concentrations in various types of waters, *Environ. Sci. Technol.* 20 (1986) 341–348.
- [37] M.S. Elowitz, U. von Gunten, H.P. Kaiser, Hydroxyl radical/ozone ratios during ozonation processes. II. The effect of temperature, pH, alkalinity, and DOM properties, *Ozone Sci. Eng.* 22 (2000) 123–150.
- [38] J.W. Moffett, R.G. Zika, Reaction kinetics of hydrogen peroxide with copper and iron in seawater, *Environ. Sci. Technol.* 21 (1987) 804–810.
- [39] T.T. Nguyen, H.J. Park, J.Y. Kim, H.E. Kim, H. Lee, J. Yoon, C. Lee, Microbial inactivation by cupric ion in combination with H₂O₂: role of reactive oxidants, *Environ. Sci. Technol.* 47 (2013) 13661–13667.
- [40] J.M. Hoover, B.L. Ryland, S.S. Stahl, Mechanism of copper(I)/TEMPO-catalyzed aerobic alcohol oxidation, *J. Am. Chem. Soc.* 135 (2013) 2357–2367.
- [41] M. Luo, L. Lv, G. Deng, W. Yao, Y. Ruan, X. Li, A. Xu, The mechanism of bound hydroxyl radical formation and degradation pathway of Acid Orange II in Fenton-like Co²⁺-HCO₃⁻ system, *Appl. Catal. A* 469 (2014) 198–205.
- [42] Z. Song, N. Wang, L. Zhu, A. Huang, X. Zhao, H. Tang, Efficient oxidative degradation of triclosan by using an enhanced Fenton-like process, *Chem. Eng. J.* 198–199 (2012) 379–387.
- [43] J.C. Yu, T.Y. Kwong, Q. Luo, Z. Cai, Photocatalytic oxidation of triclosan, *Chemosphere* 65 (2006) 390–399.
- [44] S. Rafqah, P. Wong-Wah-Chung, S. Nelieu, J. Einhorn, M. Sarakha, Phototransformation of triclosan in the presence of TiO₂ in aqueous suspension: mechanistic approach, *Appl. Catal. B* 66 (2006) 119–125.
- [45] N. Stamatis, M. Antonopoulou, I. Konstantinou, Photocatalytic degradation kinetics and mechanisms of fungicide tebuconazole in aqueous TiO₂ suspensions, *Catal. Today* 252 (2015) 93–99.
- [46] N. Stamatis, M. Antonopoulou, D. Hela, I. Konstantinou, Photocatalytic degradation kinetics and mechanisms of antibacterial triclosan in aqueous TiO₂ suspensions under simulated solar irradiation, *J. Chem. Technol. Biotechnol.* 89 (2014) 1145–1154.
- [47] V.A. Sakkas, M.A. Islam, C. Stalikas, T.A. Albanis, Photocatalytic degradation using design of experiments: a review and example of the Congo red degradation, *J. Hazard. Mater.* 175 (2010) 33–44.
- [48] I. Sirés, N. Oturan, M.A. Oturan, R.M. Rodríguez, J.A. Garrido, E. Brillas, Electro-Fenton degradation of antimicrobials triclosan and triclocarban, *Electrochim. Acta* 52 (2007) 5493–5503.